



CS670: Cryptographic Techniques for Privacy Preservation

Module 4: Private Information Retrieval

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Private Information Retrieval



Downloader



Wants to watch a movie on Netflix.

However, she does not want Netflix to know what she watched.

Private Information Retrieval

Can we download a record from a remote database server, without the server learning what we downloaded?

Yes! (under certain assumptions, though)

Private Information Retrieval

Private Information Retrieval is a cryptographic technique that allows a user to download from a remote database without the database owner learning what was downloaded!

In general, the user can learn more than what they asked for.

If we have an additional condition that the user is supposed to learn only the record they requested, we refer to it as Symmetric Private Information Retrieval (SPIR).

Private Information Retrieval

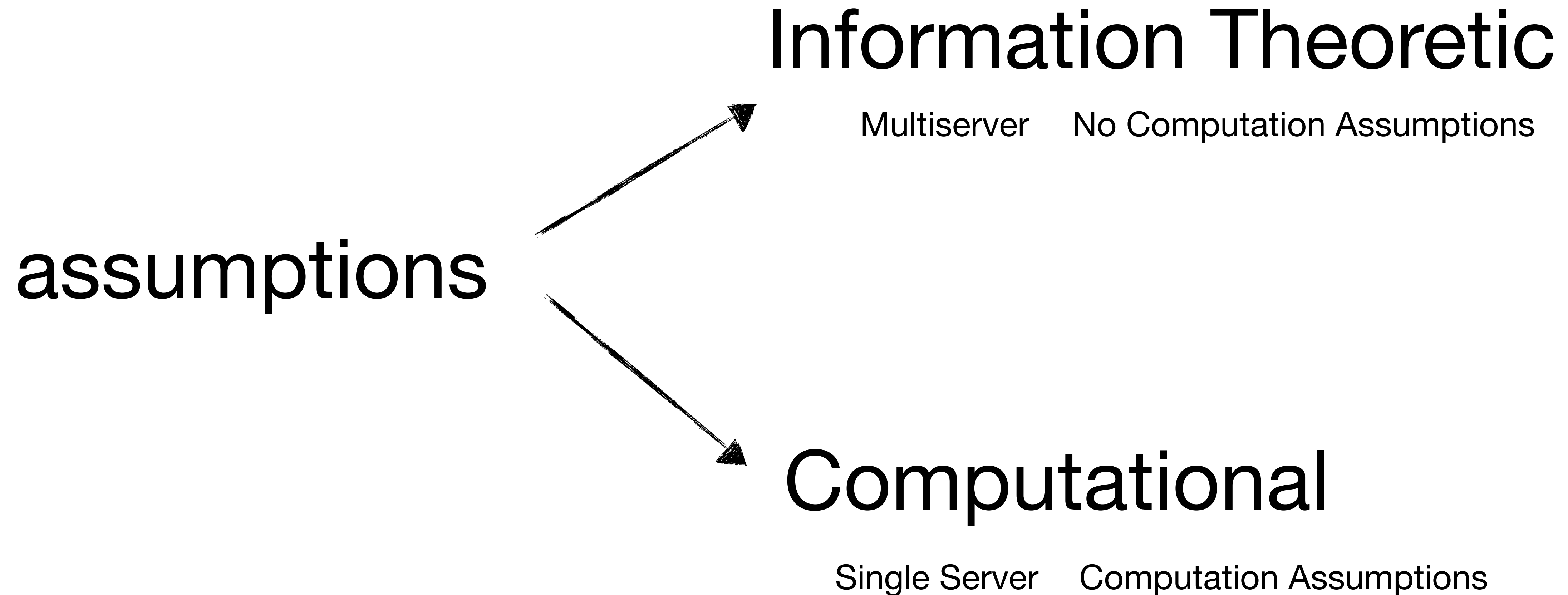
What is the difference between Private Information Retrieval and Anonymous Information Retrieval?

Tor is an example of anonymous information retrieval!

Some other examples where PIR is useful?

You may want to look up something in a patent database!

Private Information Retrieval



Simplest PIR Protocol

What is the simplest PIR protocol?

Naively downloading the entire database!



Downloader

I want to watch a video

Here is the entire Netflix Catalogue



Is this SPIR?

Private Information Retrieval

Information Theoretic Private Information Retrieval (IT-PIR)

Multiple PIR Servers

Each holding the replica of database



(a matrix of 0s and 1s)

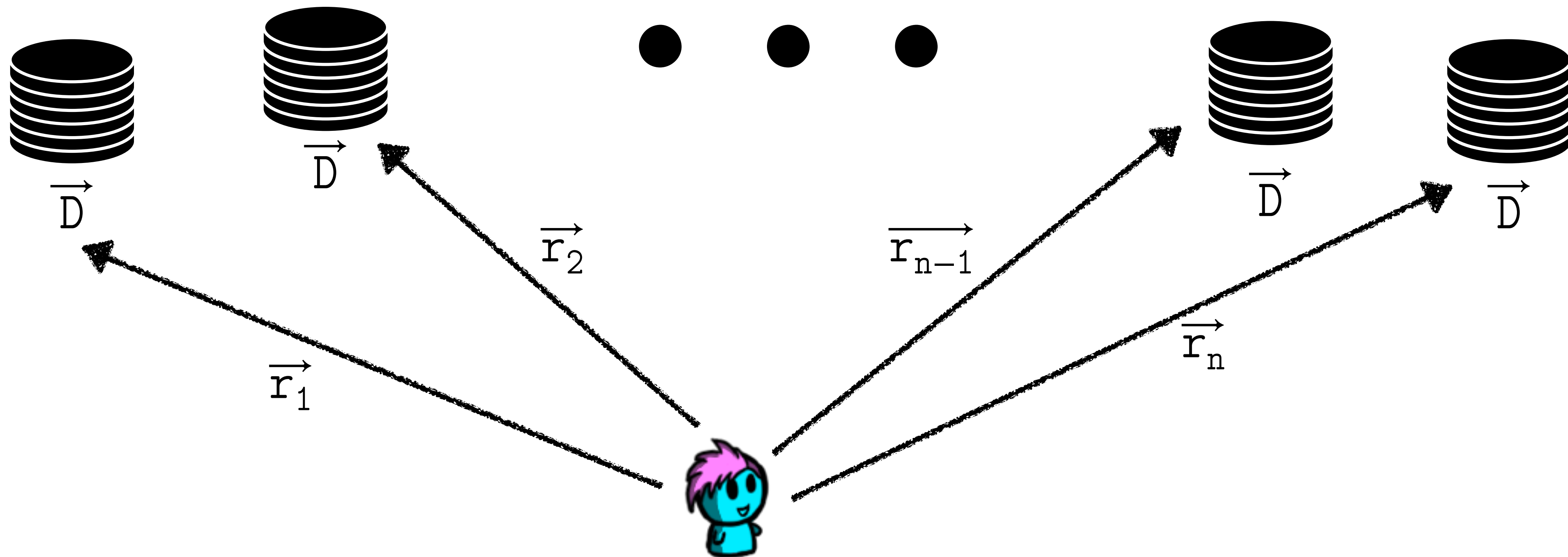
Goal: To retrieve the j^{th} row (record) of the database (matrix) privately.

IT-Private Information Retrieval

(a matrix of 0s and 1s)

Multiple PIR Servers Each holding the replica of database

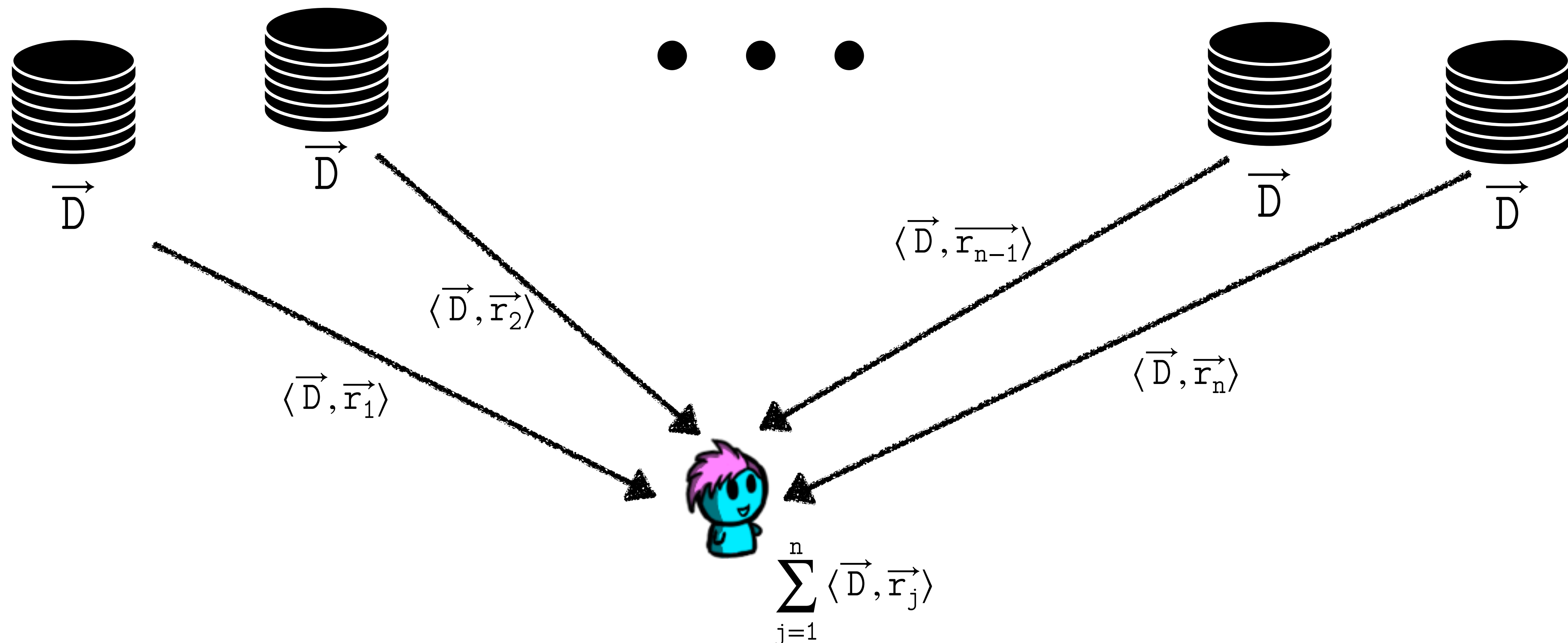
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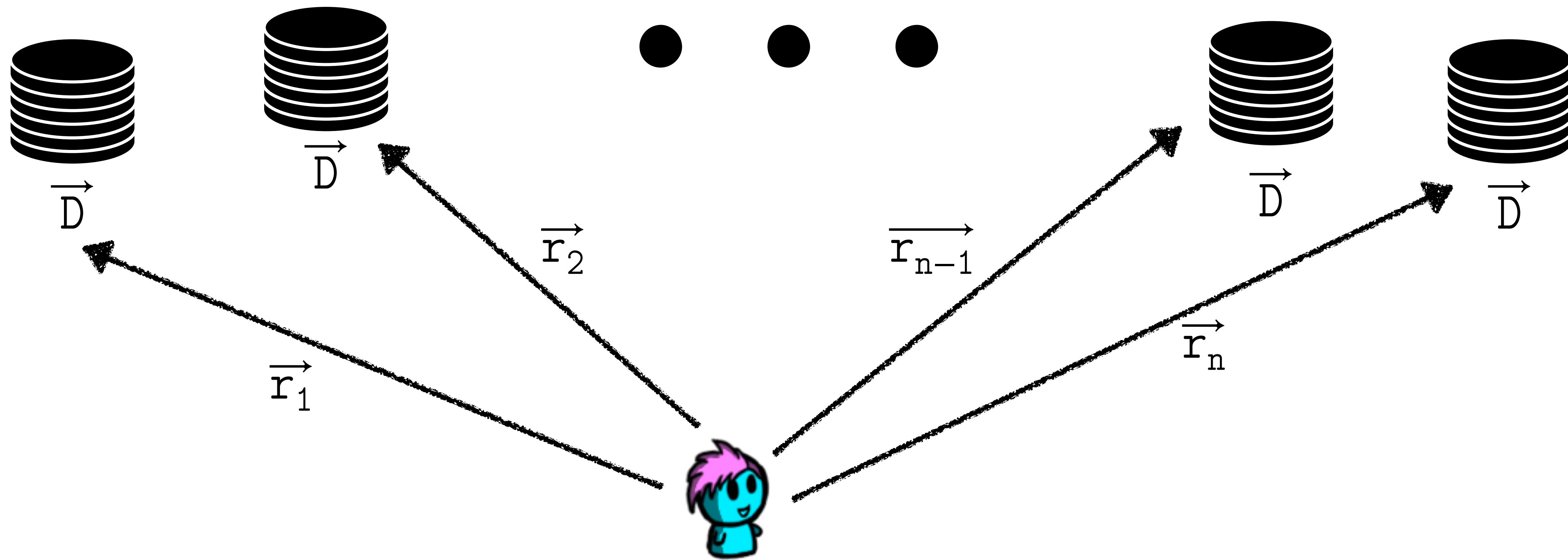
IT-Private Information Retrieval *(a matrix of 0s and 1s)*

Multiple PIR Servers Each holding the replica of database

Goal: To retrieve the j^{th} row (record) of the database (matrix) privately.



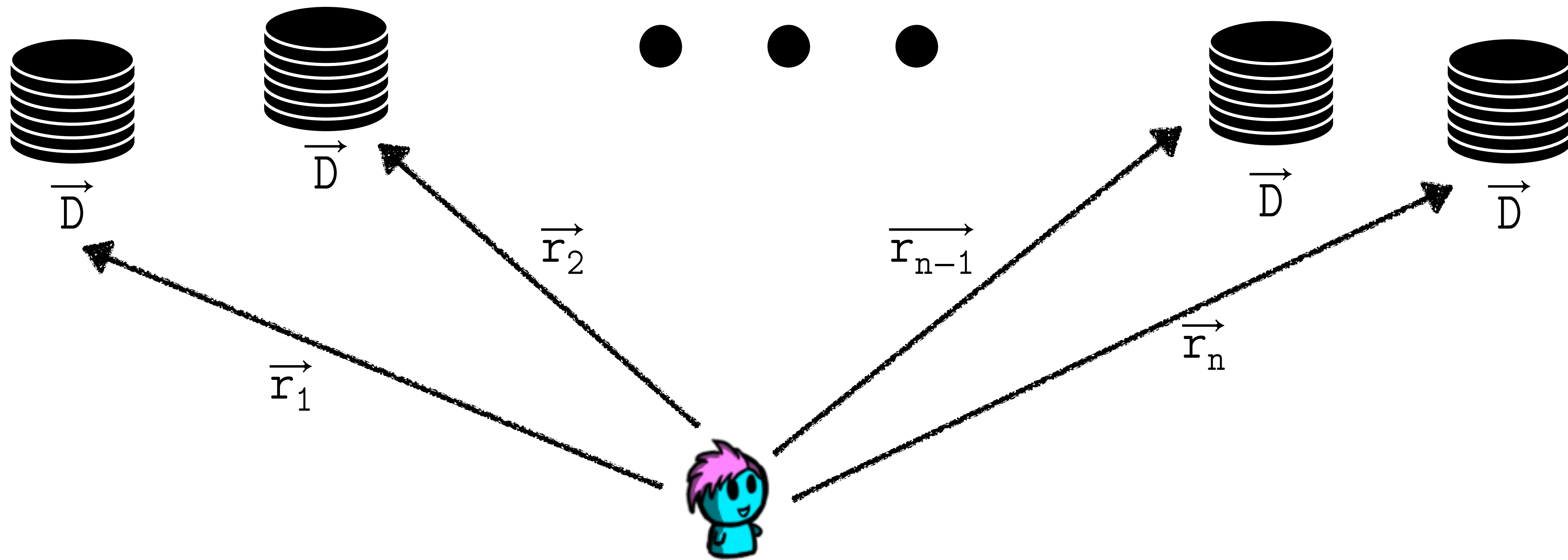
IT-Private Information Retrieval



Goal: To retrieve the j^{th} row (record) of the database (matrix) privately.

What should be the property of these vectors? $\sum_{j=1}^n \vec{r}_j = \vec{e}_i$

IT-Private Information Retrieval

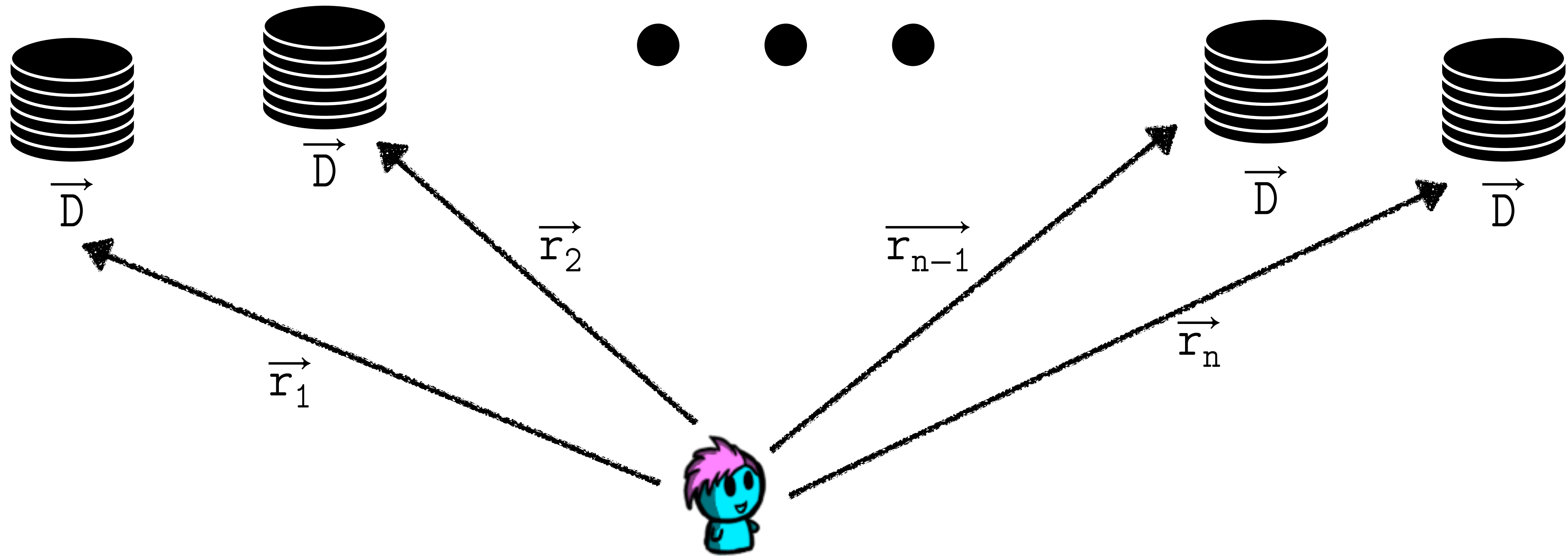


Goal: To retrieve the j^{th} row (record) of the database (matrix) privately.

What is the upload cost?

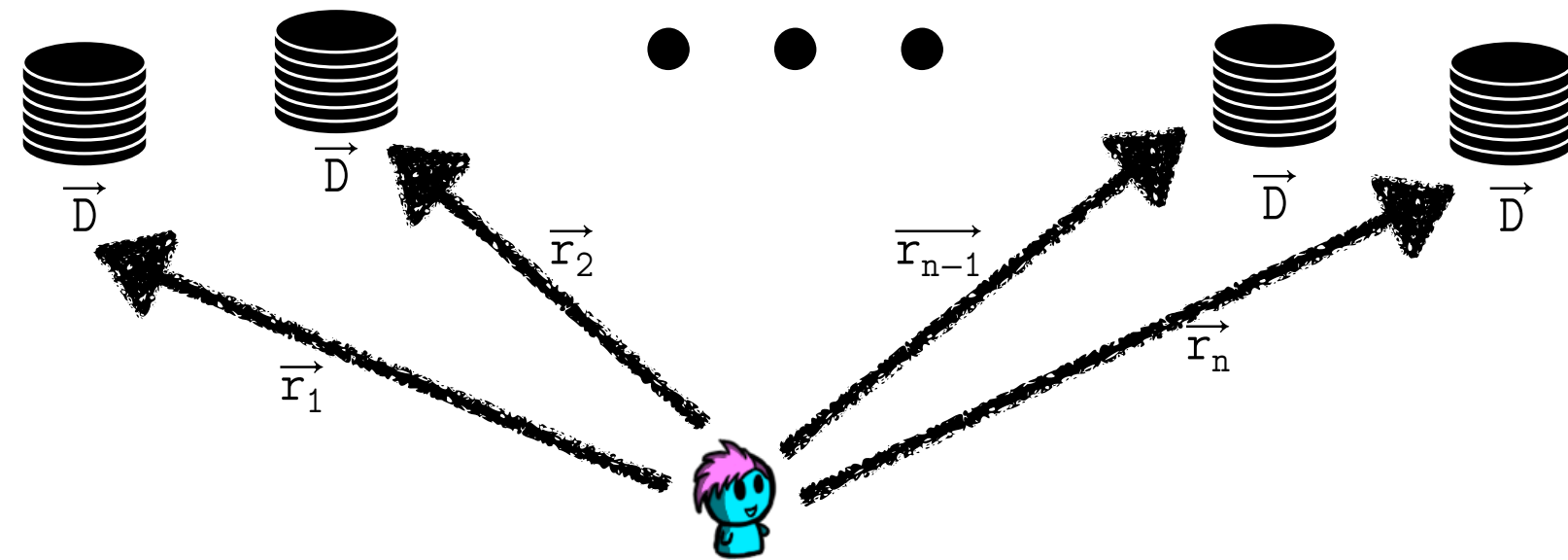
What is the download cost?

IT-Private Information Retrieval



What could go wrong?

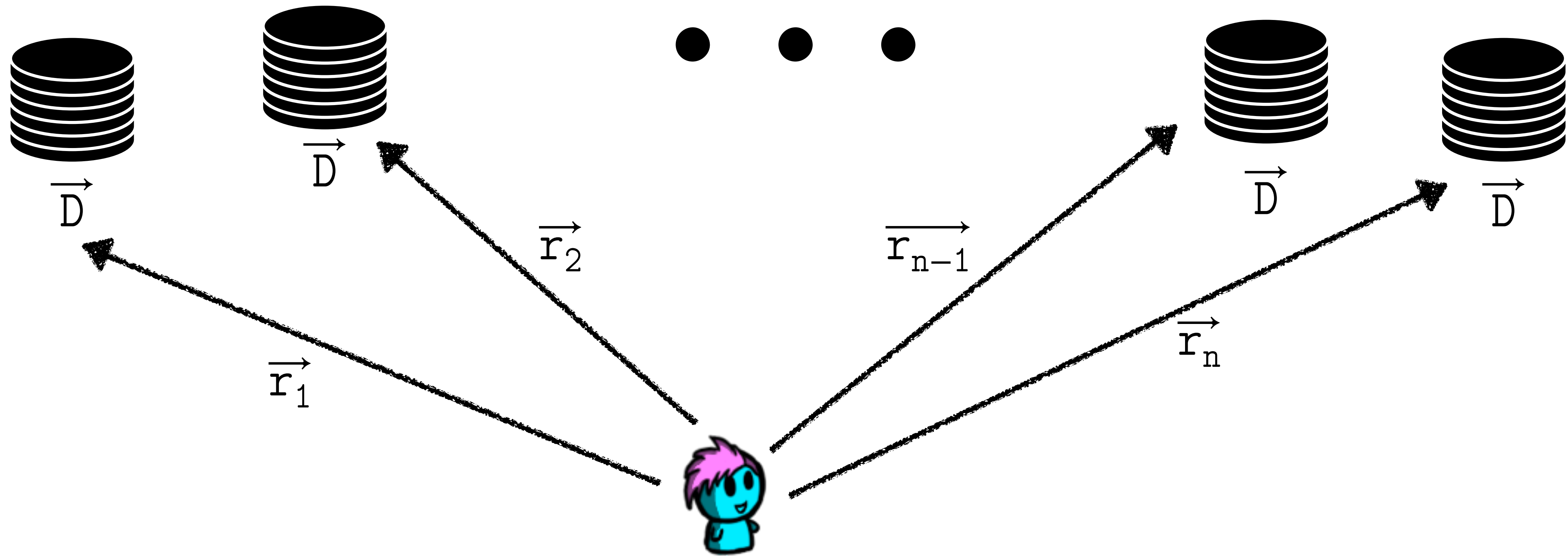
IT-Private Information Retrieval



How to protect against servers that do not respond?

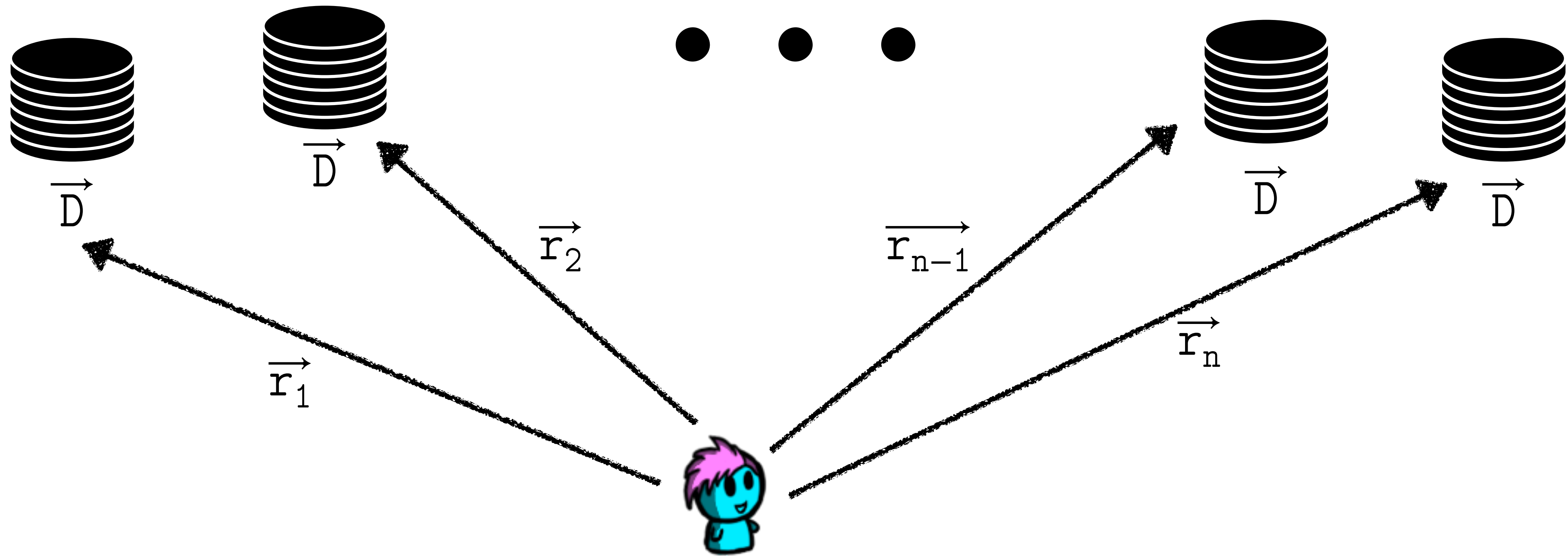
Perhaps we could use a different type of secret sharing?

IT-Private Information Retrieval



What are the downsides of this protocol?

IT-Private Information Retrieval



Can we use Distributed Point Functions here?

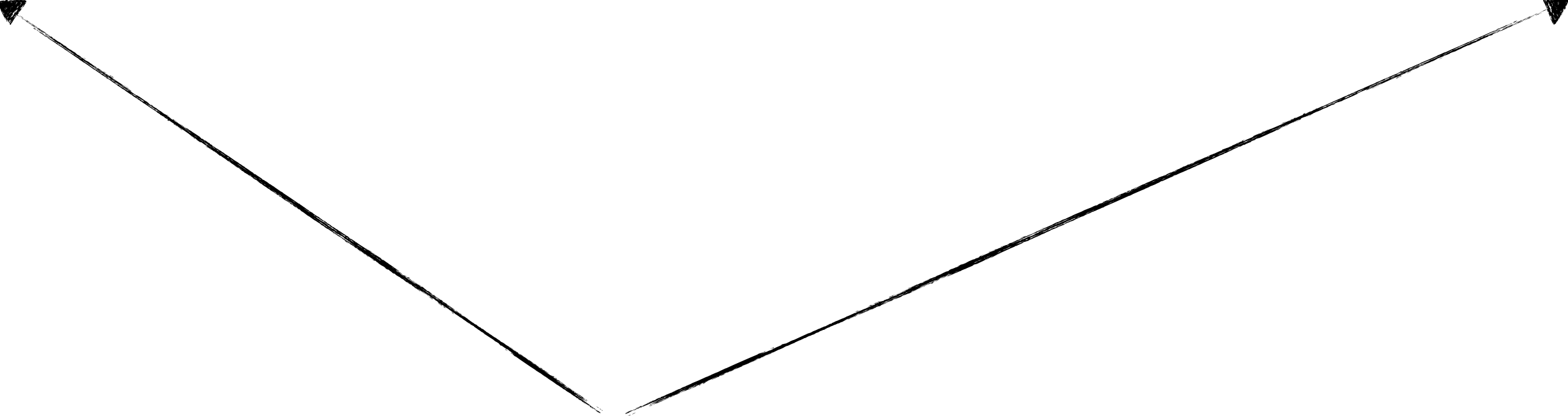
IT-Private Information Retrieval

x_{00}	x_{10}	x_{20}	x_{30}	x_{40}	x_{50}	0
x_{01}	x_{11}	x_{21}	x_{31}	x_{41}	x_{51}	0
x_{02}	x_{12}	x_{22}	x_{32}	x_{42}	x_{52}	0
x_{03}	x_{13}	x_{23}	x_{33}	x_{43}	x_{53}	0
x_{04}	x_{14}	x_{24}	x_{34}	x_{44}	x_{54}	0
x_{05}	x_{15}	x_{25}	x_{35}	x_{45}	x_{55}	0

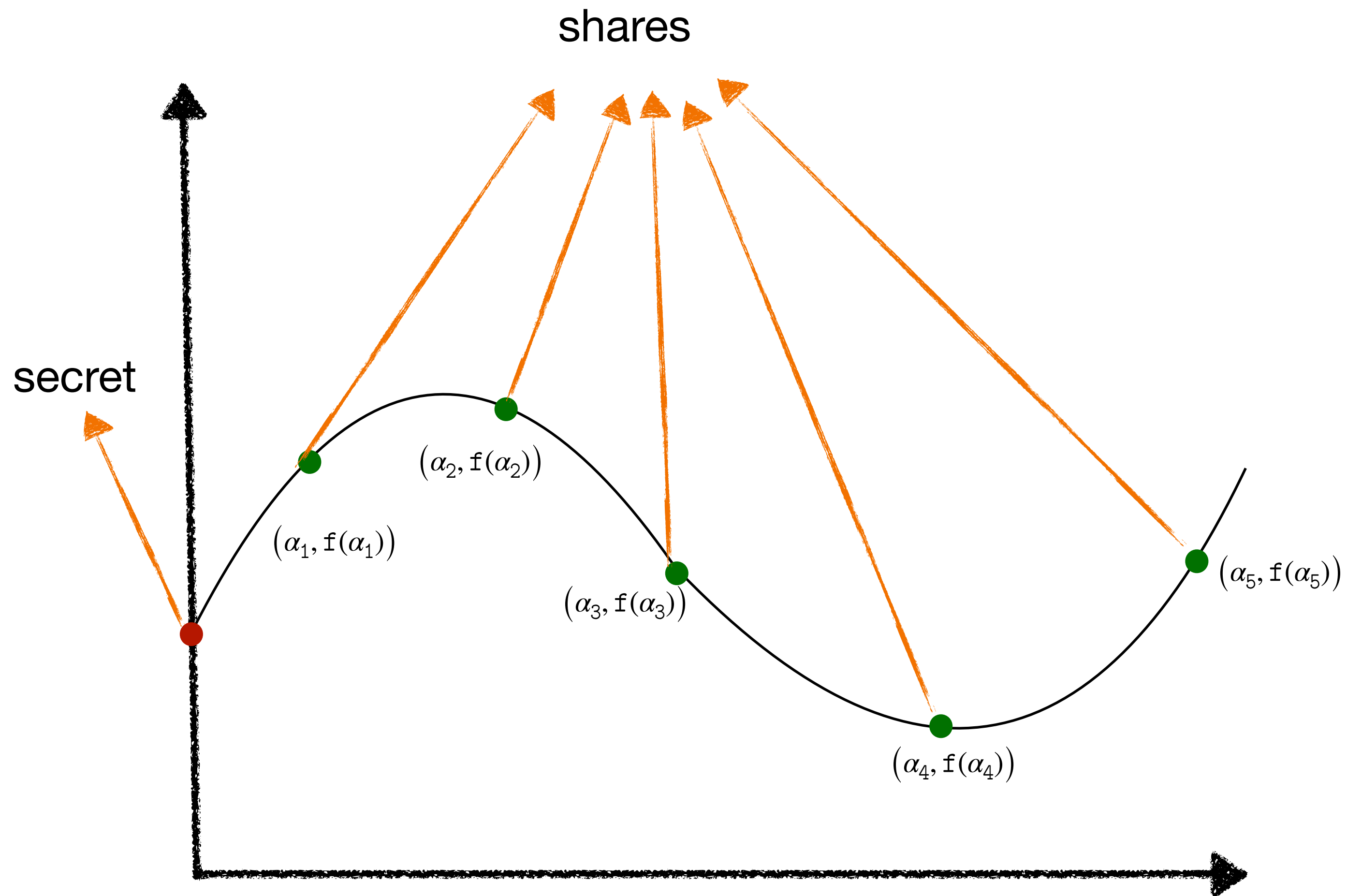
$q_1 = [0, 3, 0, 5, 3, 5]$

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x_{04}	x_{14}	x_{24}	x_{34}	x_{44}	x_{54}	0
x_{05}	x_{15}	x_{25}	x_{35}	x_{45}	x_{55}	0

$q_1 = [0, 3, 6, 5, 3, 5]$



Shamir Secret Sharing



Properties

Linearity

$$f(x) = s_1 + a_1x + a_2x^2 + \dots + a_tx^t$$

$$g(x) = s_2 + b_1x + b_2x^2 + \dots + b_tx^t$$

$$h(x) = \alpha \cdot f(x) + \beta \cdot g(x)$$

constant term is: $\alpha \cdot s_1 + \beta \cdot s_2$

each party i can locally compute:

$$h(i) = \alpha \cdot f(i) + \beta \cdot g(i)$$

Shamir Secret Sharing

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Why is linearity an important property?

What other important property does Shamir's Secret Sharing have?

Computational PIR using Fully Homomorphic Encryption

What is Fully Homomorphic Encryption?

$$(pk, sk) \leftarrow \text{KeyGen}(1^\lambda)$$

$$\text{Enc}(pk, x_0) + \text{Enc}(pk, x_1) = \text{Enc}(pk, x_0 + x_1)$$

$$\text{Enc}(pk, x_0) \cdot \text{Enc}(pk, x_1) = \text{Enc}(pk, x_0 \cdot x_1)$$

How can FHE can be used in PIR?

Keyword PIR

What do you think is Keyword-Based PIR?

We have assumed that the client knows the exact database index of the record they are looking for.

That is not always useful.

You would rather want: (key , value)